

Cerebrospinal fluid biomarkers in Parkinson disease

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Abstract | Clinical diagnosis of Parkinson disease (PD) is difficult in early stages of disease, with high risk of misdiagnosis. The long preclinical phase of PD provides the possibility for early therapeutic intervention once disease-modifying therapies have been developed, but lack of biomarkers for early diagnosis and monitoring of disease progression represents a major obstacle to achievement of this goal. Accordingly, research efforts aimed at identification of novel biomarkers have been increasing in the past 5 years. Cerebrospinal fluid (CSF) is an accessible source of brain-derived proteins, which mirror molecular changes that take place in the CNS. In this Review, we discuss evidence from numerous studies that have focused on identification of candidate CSF biomarkers for PD. Notably, molecular pathways related to α -synuclein, tau and β -amyloid peptides have received considerable attention. CSF levels of the protein DJ-1 are also of interest, although further investigation of this candidate marker is required. These studies support the usefulness of a combination of various CSF biomarkers of PD to increase diagnostic accuracy during early phases of the disease, and to differentiate PD from other neurodegenerative disorders.

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Introduction

Diagnosis of Parkinson disease (PD), a common neurodegenerative disorder, is primarily based on clinical criteria.¹ This approach, however, presents some limitations. The clinical criteria for diagnosis of PD encompass full-blown disease, but the neurodegenerative process begins several years before.¹ Early diagnosis of PD is a priority given current efforts towards development of disease-modifying therapies for this disorder.² Increasing the accuracy of PD diagnosis would also help to differentiate the disease from other forms of parkinsonism, such as multiple system atrophy (MSA), progressive supranuclear palsy (PSP) and corticobasal degeneration (CBD). PD subtypes with different clinical features and prognoses have also been proposed,³ and biomarkers to predict progression would help to differentiate between these subtypes, which could inform clinical management of the disease.

An ideal biomarker should be sensitive, reproducible, closely associated with disease pathophysiology, easy to measure, inexpensive, noninvasive and thoroughly validated.⁴ Detection of reliable biomarkers for PD is under intensive investigation. As molecular changes in the

brain are reflected in cerebrospinal fluid (CSF) composition, the CSF represents an optimal source of biomarkers of neurodegenerative diseases.⁴

Studies in Alzheimer disease (AD) have highlighted the utility of CSF biomarkers in early diagnosis. For example, amyloid- β_{1-42} ($A\beta_{42}$), total tau (t-tau) and phosphorylated tau (p-tau) are 'state markers' of the disease, in that they reliably reflect AD pathology.⁵ The prodromal phase of the disease—when the brain is already affected by the degenerative processes but the clinical picture does not yet fulfil the criteria for dementia—can be diagnosed through detection of CSF biomarkers. This knowledge has been translated to operational diagnostic criteria.^{5,6} Such a prodromal phase is characteristic of other neurodegenerative diseases, including PD, highlighting the need to follow the same route to biomarker development for early diagnosis.

With respect to CSF biomarkers of PD,^{4,7} α -synuclein (α -syn)-related biomarkers and the classical CSF biomarkers of AD have received considerable interest. As multiple disease processes might coexist in PD, dynamically influencing each other over the disease course (Figure 1), combination of different biomarkers that reflect each contributing pathogenic mechanism is likely to be the most appropriate approach.⁸ For example, tau and α -syn pathology coexist in PD, as well as in AD and dementia with Lewy bodies (DLB).^{9,10} α -Syn induces aggregation and polymerization of tau, which promotes formation of abundant intracellular amyloid-tau inclusions.¹¹ These dynamic interactions explain the overlap of both pathological and clinical features that are frequently observed

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Competing interests

A. Castrioto declares an association with the following organization: Novartis. P. Calabresi declares associations with the following companies: Bayer Schering, Biogen, Boehringer Ingelheim, Lundbeck, Sanofi-Aventis, Sigma-Tau. See the article online for full details of the relationships. P. Calabresi, L. Parnetti and O. El-Agnaf have submitted a patent application (UK 1011420.5) for a method for Parkinson disease diagnosis. The other authors declare no competing interests.

Key points

- Clinical diagnosis of Parkinson disease (PD) can be difficult at early stages of the disease course
- Biomarkers for diagnosis and prognosis of PD are not currently available
- Cerebrospinal fluid levels of α -synuclein, DJ-1, amyloid- β_{1-42} , tau and lysosomal enzymes are promising candidates for PD biomarkers
- When used alone, these biomarkers have low specificity and sensitivity for PD
- Combination of several biomarkers that reflect different pathogenic pathways is necessary

in patients with tauopathies and synucleinopathies.¹² Moreover, co-occurrence of tau and α -syn pathology has been detected in neurons and oligodendrocytes in various neurodegenerative diseases, such as AD, PD and DLB.¹³

Several lines of evidence support a pathogenic link between lysosomal dysfunction and PD,^{14–19} suggesting measurement of lysosome activity as a biomarker of this disease. In addition, CSF measurement of DJ-1—a multifunctional protein implicated in the response to oxidative stress during neurodegeneration—has also received attention.²⁰

In this article, we review the potential use of CSF proteins as biomarkers in PD, focusing on α -syn, DJ-1, AD-associated biomarkers (A β_{42} , t-tau and p-tau) and lysosomal enzymes.

α -Synuclein

Role in synucleinopathies

α -Syn is a 140 amino acid member of a highly conserved family of proteins,²¹ which localize predominantly at presynaptic neuronal terminals under physiological conditions.²² The precise function of α -syn is not known, but its localization and ability to interact with membranes and lipids suggests a role in synaptic plasticity, brain lipid metabolism, and regulation of vesicle dynamics and trafficking at presynaptic terminals.^{13,23} α -Syn is the main component of Lewy bodies, which are a hallmark of synucleinopathies.²⁴ Lewy bodies primarily consist of amyloid-like fibrils composed of post-translationally modified forms of α -syn.²⁵ α -Syn is also found in glial cytoplasmic inclusions in MSA.²⁶

The importance of α -syn in PD pathogenesis was recognized when a missense mutation (Ala53Thr) in the α -syn gene (SNCA) was linked to a rare, dominant form of PD.²⁷ Two further point mutations in SNCA (Ala30Pro and Glu46Lys) have been identified in other families with PD.^{28,29} Triplication of the SNCA locus is linked with young-onset cases of severe PD with some features of DLB,³⁰ and SNCA duplication causes familial PD with a similar phenotype to late-onset PD.³¹ Therefore, elevated expression of wild-type α -syn increases the risk of developing PD, and α -syn protein levels may be an important determinant of the severity of the parkinsonian phenotype.¹³ Population studies have also demonstrated that SNCA polymorphisms are associated with increased risk of sporadic PD and MSA, which reinforces the role of α -syn in the pathogenesis of synucleinopathies.^{32,33}

α -Syn aggregation is a key event in the pathogenesis of synucleinopathies. The morphology of wild-type or mutant α -syn fibrils that are generated *in vitro* is similar

to fibrils in Lewy bodies.^{34,35} α -Syn fibrillation (Figure 1) involves α -syn oligomerization followed by oligomer conversion into mature amyloid fibrils,³⁶ which are toxic to cultured neuronal cells.³⁷ Collectively, these data suggest a central role for α -syn deposition in the pathogenesis of synucleinopathies.

A biomarker for PD?

Several recent studies have explored the potential use of α -syn as a putative PD biomarker in human biological fluids (Table 1). α -Syn was first detected in CSF using immunoprecipitation and immunoblotting with α -syn-specific antibodies.³⁸ Owing to the low sensitivity of these methodologies, however, no differences between controls and PD were observed.

Retrospective studies have investigated the levels of total α -syn (t- α -syn) in CSF as measured with ELISA, but have produced inconclusive results: some studies have shown a clear trend of lower CSF t- α -syn in PD and other synucleinopathies such as MSA and DLB,^{20,39–42} whereas in other investigations α -syn levels in PD largely overlap with those in different diseases.^{8,43–46} In these studies, CSF levels of t- α -syn could not be used to discriminate between patients with synucleinopathies and cognitively normal individuals with other, non-neurodegenerative diseases. The use of t- α -syn as CSF biomarker for PD, therefore, was not confirmed.

Many factors might contribute to the discordance of the above results, including variability in patients' characteristics, such as age, disease duration and disease severity. Another limitation is heterogeneity of control groups, which comprised healthy individuals^{44,45} and patients with no cognitive dysfunction who had either non-neurological or neurological pathology.^{8,39,40,42,44,45,47} Inconsistent results might also be caused by methodological issues, including pre-analytical differences in collection, handling and storage of CSF samples, use of antibodies that detect different species of α -syn, and variability among sources of recombinant α -syn protein used as the standard in ELISA assays. Another critical factor is represented by blood contamination, which could cause substantial increases in CSF α -syn concentration.²⁰

As discussed above, α -syn and tau interact dynamically and synergistically to promote mutual aggregation. A reasonable strategy in the context of detecting CSF markers of PD, therefore, is simultaneous measurement of these misfolded proteins, as demonstrated by two recent studies (Table 2).^{8,40} In one study,⁴⁰ CSF t- α -syn and t-tau levels were measured in 721 patients with the synucleinopathies PD, DLB and MSA, and the tauopathies AD and PSP. Combination of CSF t- α -syn and t-tau values provided the most-accurate discrimination of synucleinopathies from the other neurological disorders ($P < 0.0001$).⁴⁰ In the other study, lower t-tau:t- α -syn and p-tau:t- α -syn ratios discriminated PD from DLB, FTD and AD (Supplementary Table 1 online).⁸

Oligomeric and phosphorylated α -syn

Substantial evidence suggests that soluble oligomers of amyloid proteins are associated with neurodegenerative

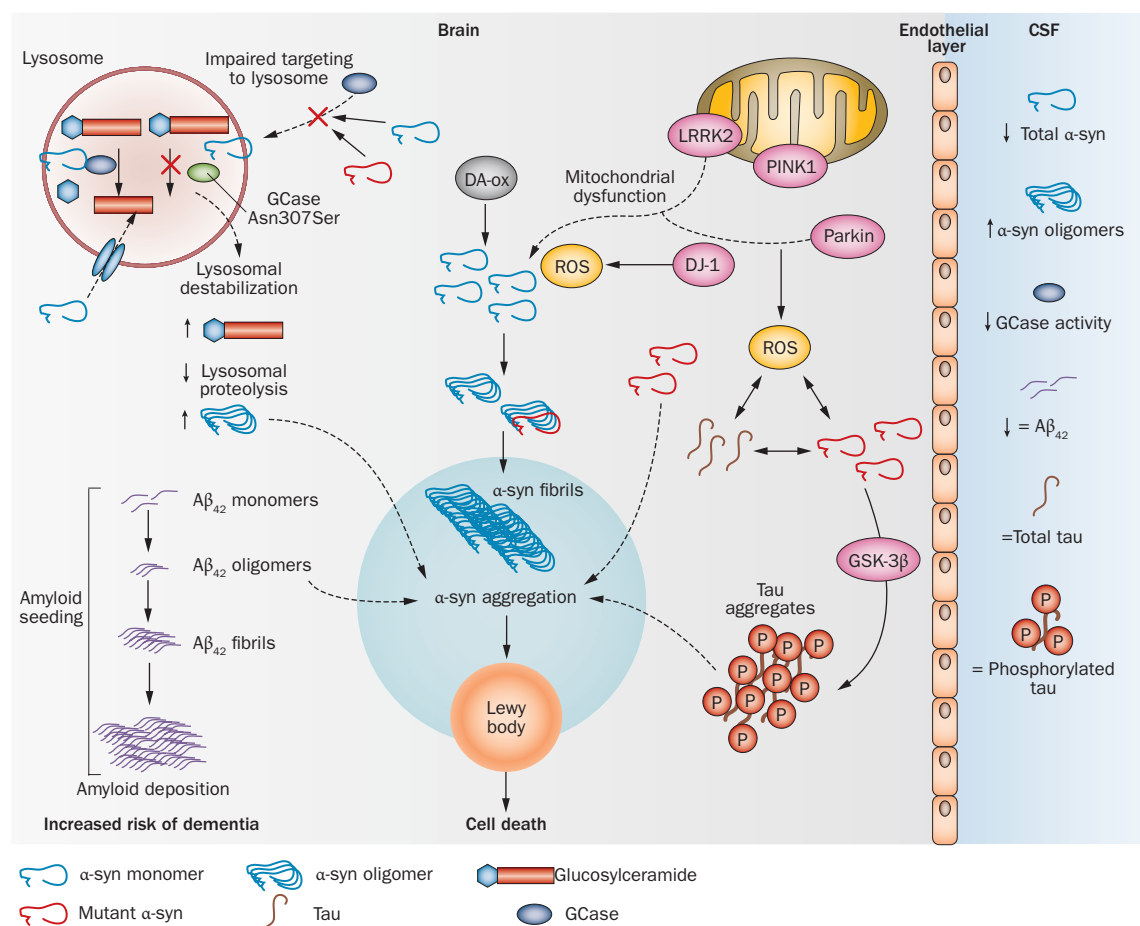


Figure 1 | Putative pathogenic pathways underlying CSF biomarkers of PD. Aggregation of α -syn into oligomers, fibrils and then Lewy bodies—the hallmark of PD—is triggered by factors such as DA-ox, or ROS released following mitochondrial dysfunction. Loss of function of LRRK2, PINK1, Parkin and DJ-1, can cause or exacerbate mitochondrial function. ROS also interact with α -syn and tau, increasing formation of hyperphosphorylated tau filaments, which aggregate into Lewy bodies. α -Syn impairs lysosomal targeting of GCase, decreasing lysosomal proteolysis and functionality, and promoting α -syn aggregation. This effect is amplified by mutations in GCase (Asn370Ser). As a consequence of these pathogenic pathways, CSF levels of total α -syn and GCase activity are reduced in PD, and CSF levels of α -syn oligomers are increased. Cognitive dysfunction and dementia in PD might be linked to aggregation of amyloid peptides (mainly $A\beta_{42}$) into oligomers and fibrils, which boosts α -syn aggregation and might result in decreased CSF levels of $A\beta_{42}$. Abbreviations: A β , amyloid- β ; DA-ox, oxidatively modified dopamine; GCase, glucocerebrosidase; GSK-3 β , glycogen synthase kinase 3 β ; LRRK2, leucine-rich repeat kinase protein 2; PD, Parkinson disease; PINK1, PTEN-induced putative kinase protein 1; ROS, reactive oxygen species; α -syn, α -synuclein.

diseases, and these oligomers may be the pathogenic species that cause neuronal cell death.³⁷ Soluble oligomers of α -syn (o- α -syn) are neurotoxic *in vitro* and *in vivo*.^{48,49} Levels of these species are elevated in brain homogenates in PD and DLB compared with normal brains,^{50,51} suggesting a role for o- α -syn in PD pathogenesis. Accordingly, detection of these species in CSF could represent a good biomarker of PD and related synucleinopathies.⁵² In line with this premise, CSF o- α -syn levels and o- α -syn:total- α -syn ratios were found to be substantially higher in patients with PD—including those with mild and early-stage disease—than in individuals with other neurological disorders, or those who underwent lumbar puncture for diagnostic purposes.⁵³ CSF o- α -syn had a sensitivity of 75.0% and a specificity of 87.5% for PD, and the CSF o- α -syn:t- α -syn ratio had a sensitivity of 89.3% and a specificity

of 90.6%. These findings have been confirmed in two independent reports.^{54,55}

Lewy bodies are formed mostly of post-translationally modified forms of α -syn. The modifications include phosphorylation, nitration and ubiquitination, and most of the accumulated α -syn in Lewy bodies is phosphorylated at residue Ser129 (p129- α -syn). Although some studies support a pathogenic role for phosphorylation of Ser129, whether p129- α -syn promotes or protects against PD is unclear. Postmortem analysis of o- α -syn and monomers and oligomers of p129- α -syn in samples of ventricular CSF from patients with MSA, PD and DLB revealed elevated levels of oligomeric p129- α -syn in patients with MSA compared with in other groups.⁵⁶ p129- α -Syn and t- α -syn levels were also measured in fresh CSF samples from a large cohort (~600 samples) of patients with various types of parkinsonism—including

Table 1 | Studies on cerebrospinal fluid levels of α -synuclein

Study	Patients	Analytes measured	Outcome	Correlation with analyte levels		
				Age	Disease duration	Disease severity
Borghi <i>et al.</i> (2000) ^{38*}	12 PD, 10 controls	t- α -syn	No difference	ND	ND	ND
Tokuda <i>et al.</i> (2006) ⁴¹	33 PD, 38 controls	t- α -syn	↓ in PD	Inverse	ND	Inverse (H&Y stage)
Mollenhauer <i>et al.</i> (2008) ³⁹	8 PD, 38 DLB, 13 AD, 8 CJD, 13 controls	t- α -syn	↓ in PD and DLB, ↑ in CJD	Inverse	ND	ND
Ohrfelt <i>et al.</i> (2009) ⁴⁴	15 PD, 15 DLB, 66 AD, 55 controls	t- α -syn	↓ in AD	No	Inverse	Positive (MMSE)
Noguchi-Shinohara <i>et al.</i> (2009) ⁴³	16 DLB, 21 AD	t- α -syn	No difference	No	Inverse correlation only in DLB group	ND
Spies <i>et al.</i> (2009) ⁴⁵	40 DLB, 131 AD, 28 vascular dementia, 39 FTD	t- α -syn	No differences (trend towards inverse correlation with tau)	Inverse	ND	ND
Hong <i>et al.</i> (2010) ^{20†}	117 PD, 50 AD, 132 controls	t- α -syn	↓ in PD	Positive	ND	No
Tokuda <i>et al.</i> (2010) ⁵³	32 PD, 28 controls (t- α -syn, o- α -syn and o- α -syn: α -syn ratio); 25 PD, 35 AD, 18 PSP, 43 controls (o- α -syn)	t- α -syn o- α -syn o- α -syn: α -syn ratio	Trend towards ↓ in PD ↑ in PD ↑ in PD	No	No	No (H&Y stage)
Parnetti <i>et al.</i> (2011) ⁸	38 PD, 32 DLB, 48 AD, 31 FTD, 32 controls with other neurological disease	t- α -syn Total tau: α -syn ratio	↓ in all pathological groups (particularly in DLB and FTD) ↓ in PD versus other pathological groups	No	No	ND
Mollenhauer <i>et al.</i> (2011) ⁴⁰	Cohort 1: 62 AD, 55 DLB, 51 PD, 29 AD, 76 controls Cohort 2: 22 NPH, 8 PSP, 66 DLB, 273 PD, 15 MSA, 23 controls Cohort 3: 21 AD, 13 DLB, 7 controls	t- α -syn	↓ in PD, DLB, MSA Combination of age, α -syn and t-tau provides best discrimination of synucleinopathies	ND	ND	No
Shi <i>et al.</i> (2011) ^{63‡}	126 PD, 32 MSA, 50 AD, 137 controls	t- α -syn	↓ in PD compared with controls and AD	ND	ND	No
Aerts <i>et al.</i> (2012) ⁴⁶	58 PD, 47 MSA, 3 DLB, 22 vascular parkinsonism, 10 PSP, 2 corticobasal degeneration	t- α -syn	No difference	ND	ND	ND
Foulds <i>et al.</i> (2011) ⁵⁶	38 PD, 16 DLB, 12 PSP, 8 MSA, 20 controls Postmortem cerebrospinal fluid	t- α -syn, p- α -syn Phosphorylated o- α -syn	No difference ↑ in MSA	No	No	No
Tateno <i>et al.</i> (2012) ⁴²	11 PD, 6 DLB, 11 MSA, 9 AD, 11 controls	t- α -syn	↑ in AD ↓ in PD, DLB, MSA (no difference among these groups)	ND	ND	ND
Park <i>et al.</i> (2011) ⁵⁴	23 PD, 29 controls	t- α -syn o- α -syn	No difference ↑ in PD	No	No	ND
Wang <i>et al.</i> (2012) ⁵⁷	Discovery: 93 PD, 16 MSA, 33 PSP, 26 AD, 78 controls Validation: 116 PD, 25 MSA, 27 PSP, 50 AD, 126 controls	t- α -syn p- α -syn p- α -syn: α -syn ratio	↓ in PD and MSA ↑ in PD ↑ in PD	ND	No	Positive (UPDRS), no correlation with H&Y stage

Unless otherwise stated, results were obtained using ELISA analysis. *Study performed by means of immunoprecipitation and immunoblotting. ‡Study performed using xMAP analysis. Abbreviations: AD, Alzheimer disease; CJD: Creutzfeldt–Jacob disease; DLB, dementia with Lewy bodies; H&Y Hoehn and Yahr; MMSE, Mini-Mental State Examination; MSA, multiple system atrophy; ND, not determined; NPH, normal pressure hydrocephalus; o- α -syn, oligomeric α -synuclein; PD, Parkinson disease; PSP, progressive supranuclear palsy; p- α -syn, phosphorylated α -synuclein; t- α -syn, total α -synuclein; UPDRS, Unified Parkinson Disease Rating Scale.

PD, MSA and PSP—as well as healthy individuals.⁵⁷ CSF p129- α -syn levels showed a positive correlation with PD severity, and combination of CSF t- α -syn and p129- α -syn improved discrimination between PD and other forms of parkinsonism.⁵⁷

Some researchers have postulated that native α -syn has an α -helically folded tetrameric structure, which ensures stability and resistance to self-aggregation.^{58,59} At present,

available immunoassays cannot distinguish between toxic oligomers and tetramers. A subsequent study, however, has challenged the α -syn tetramer hypothesis by demonstrating that unstructured monomers of α -syn represent the predominant form in native conditions.⁶⁰ Further research is needed to clarify these conflicting reports, the results of which would have implications for biomarker development.

Table 2 | Studies on classical cerebrospinal fluid AD biomarkers in PD

Study	CSF A β_{42}	CSF total tau	CSF phospho-tau	Comments
Sjogren <i>et al.</i> (2002) ⁸¹	↓ in AD and ALS, ↓ in FTD and PD	↑ in AD	↑ in AD	–
Bibl <i>et al.</i> (2006) ⁹²	A β_{42} /A β_{37} ↓ in AD versus controls A β_{1-40} ↑ in DLB versus PDD	ND	ND	–
Mollenhauer <i>et al.</i> (2006) ⁷⁸	↓ in PDD versus PD	↑ in PDD versus PD	ND	No correlations with age, disease duration, H&Y stage or MMSE score
Mollenhauer <i>et al.</i> (2007) ⁸⁶	↓ in AD, PSP, CBD and DLB	↑ in AD Slight ↑ in PD and CBD versus controls	ND	Positive correlation with MMSE score
Parnetti <i>et al.</i> (2008) ⁸⁰	↓ in DLB > PDD > PD compared with controls	DLB > PDD > PD No differences between PD and controls	No difference	↓ A β_{42} associated with ↓ performance in word fluency tests and prose memory tests in PD
Zhang <i>et al.</i> (2008) ⁸⁷	↓ in PD	↑ in AD	ND	–
Compta <i>et al.</i> (2009) ⁹³	↓ in PDD > PD compared with controls	↑ in PDD versus PD and controls	↑ in PDD versus PD and controls	↓ A β_{42} associated with ↓ verbal fluency; ↑ tau associated with worse memory, visuospatial functions and naming, and with increasing age
Siderowf <i>et al.</i> (2010) ⁸⁴	Unchanged	Unchanged	Unchanged	↓ baseline A β_{42} associated with faster cognitive decline (↑ A β_{42} independent predictor of cognitive decline)
Alves <i>et al.</i> (2010) ⁸²	↓ in PD	No difference	No difference	↓ A β_{42} associated with memory impairment
Montine <i>et al.</i> (2010) ⁷⁹	↓ in PD-CIND and PDD versus PD and controls	No difference	No difference	–
Příkrylová <i>et al.</i> (2010) ⁸⁸	No difference between groups	↑ in PD	ND	↑ clusterin levels; t-tau, t-tau:A β_{42} ratio and clusterin levels associated with disease duration
Leverenz <i>et al.</i> (2011) ⁹⁴	Unchanged	Unchanged	ND	↓ A β_{42} and/or A β_{42} :t-tau ratio associated with ↑ performance in digit symbol tests and category fluency tests
Parnetti <i>et al.</i> (2011) ⁸	No difference between PD and controls	No difference between PD and controls	No difference between PD and controls	t-tau and p-tau associated with age and disease duration; A β_{42} associated with age
Shi <i>et al.</i> (2011) ⁶³	Slight ↓ in PD and MSA versus controls	↓ in PD versus controls	↓ in PD versus controls	Fractalkine:A β_{42} ratio associated with H&Y stage & UPDRS score; t-tau and p-tau associated with age
Andersson <i>et al.</i> (2011) ⁸⁹	↓ in DLB versus PDD	↑ in DLB, ↑ in PDD	No difference	–
Příkrylová <i>et al.</i> (2011) ⁸³	Slight ↓ in NT PD	↑ tau and tau:A β ratio in NT PD	ND	Tau levels associated with disease severity
Bech <i>et al.</i> (2011) ⁹⁰	↓ in DLB versus others groups	No difference	No difference	–

Abbreviations: A β_{42} , amyloid- β_{1-42} peptide; AD, Alzheimer disease; ALS, amyotrophic lateral sclerosis; aMCI, amnesic mild cognitive impairment; CBD, corticobasal degeneration; DLB, dementia with Lewy bodies; FTD, frontotemporal dementia; H&Y, Hoehn and Yahr; MMSE, Mini-Mental State Examination; MSA, multiple system atrophy; NT, non-tremor dominant; PD, Parkinson disease; PD-CIND, PD with cognitive impairment not demented; PDD, PD-dementia; PSP, progressive supranuclear palsy; p-tau, phosphorylated tau; TD, tremor dominant; T-tau, total tau; UPDRS, Unified Parkinson Disease Rating Scale.

DJ-1

DJ-1 is the product of *PARK7*—a gene that is linked to familial PD—and has a protective role in oxidative stress during neurodegeneration.⁶¹ The level of DJ-1 in the CSF has recently been proposed as a potential PD biomarker, although the data are conflicting. CSF DJ-1 levels (measured with quantitative immunoblotting) were considerably higher in individuals with early-stage PD compared with age-matched controls.⁶² However,

a study in a larger cohort using multiplex technology, which allows simultaneous measurement of several molecules in a very small volume of biological fluid with high sensitivity and specificity, provided different results, showing lower DJ-1 levels in PD compared with controls, a positive correlation between DJ-1 levels and patient age, and no correlation with disease severity.²⁰ In another study, CSF DJ-1 levels in PD were lower than in healthy controls and AD, but were not distinguishable

from those in the MSA group.⁶³ The variability of these results might be due, in part, to methodological issues, such as differences in techniques and assays used, sample sizes, and composition of the control group.

Striatal dopaminergic degeneration is a key pathological process in PD. This process was assessed using PET in combination with CSF DJ-1 and t- α -syn determination in eight symptomatic and 18 asymptomatic carriers of mutated leucine-rich repeat kinase 2, which causes some forms of familial PD.⁶⁴ The data revealed no correlation between the loss of striatal dopaminergic function and CSF levels of DJ-1 and t- α -syn, suggesting that these biomarkers on their own do not reflect disease progression.⁶⁴

AD-associated CSF biomarkers

PD is not only a movement disorder. Cognitive impairment is more common in this condition than in the general population, affecting approximately 30% of patients, and occurring even during early stages of the disease.^{65,66} The prevalence of dementia increases up to 80% in patients with more than 20 years of disease duration.⁶⁷ Cognitive impairment and dementia have a dramatic effect on quality of life of patients and their caregivers.

Neuropathological studies have repeatedly highlighted the co-occurrence of diffuse Lewy body pathology and amyloid plaques in neocortical regions in patients with PD dementia (PDD).^{68–70} Senile plaques are the result of abnormal processing of amyloid precursor protein, with subsequent misfolding and aggregation of A β peptides.²⁴ Double transgenic mice with neuronal expression of human A β peptides and α -syn exhibit functional and morphological alterations that resemble the Lewy body variant of AD⁷¹—that is, the AD variant characterized clinically by the presence of extrapyramidal signs and neuropathologically by coexistence of AD and Lewy body pathology. In these mice, α -syn aggregation is potentiated by A β ₄₂, and accumulation of A β ₄₂ and tau is exacerbated by α -syn.^{72,73} Moreover, A β peptides can contribute to development of Lewy body diseases through promotion of α -syn aggregation. These data support the role of AD pathology in the pathogenesis of cognitive decline in PD.^{66,74,75}

The high risk of cognitive impairment in PD calls for a search for biomarkers that can predict dementia in this disease. Many studies have focused on classical AD CSF biomarkers, such as A β ₄₂, t-tau and p-tau (Table 2). Increased levels of t-tau and p-tau, together with a decrease in A β ₄₂, characterize the AD CSF profile.⁷⁶ Moreover, a recent longitudinal study in dominantly inherited AD has confirmed that CSF biomarkers are altered many years before the clinical onset of disease.⁷⁷ Most studies in patients with PD have reported a slight reduction in A β ₄₂^{63,78–84} and no changes in t-tau and p-tau levels.^{80,82,85–90} Particularly low CSF A β ₄₂ levels are observed in patients with AD, DLB or PDD compared with controls and patients with PD.^{80,91,92}

The link between CSF A β ₄₂ levels and cognitive decline in PD has been studied extensively.^{79,82,84,93,94} Compta

*et al.*⁹³ found lower levels of CSF A β ₄₂ and increased t-tau and p-tau levels in PDD compared with controls and PD without dementia. In patients with PDD, higher concentrations of CSF t-tau and p-tau were associated with greater impairment of memory and of naming and visuospatial abilities, whereas reduced A β ₄₂ levels correlated with impaired phonetic and semantic verbal fluency.⁹³ CSF concentrations of A β ₄₂, and the less-aggregation-prone A β ₄₀ and A β ₃₈, as well as t-tau and p-tau were measured in 109 patients with *de novo* PD without dementia, and in age-matched controls and patients with mild AD.⁸² CSF A β ₄₂, A β ₄₀ and A β ₃₈ levels were reduced in patients with PD compared with controls, and correlated negatively with memory impairment. Similarly, lower A β ₄₂ levels and A β ₄₂:tau ratio in the CSF were associated with worse performance on the Digit Symbol Test in PD patients without dementia.⁹⁴

A prospective study in patients with PD showed that lower CSF levels of A β ₄₂ at baseline were associated with more-rapid cognitive decline, particularly in tests of attention, conceptualization and memory.⁸⁴ This study provided direct evidence that low CSF A β ₄₂ levels might predict cognitive decline in patients with PD patients.

The burden of cortical A β deposition is reliably reflected by PET amyloid-specific radiotracers, such as ¹¹C-labelled Pittsburgh compound B (¹¹C-PiB), and increased cortical amyloid burden is typically observed in patients with AD.^{95,96} In patients with PDD, elevated ¹¹C-PiB cortical uptake has been described, consistent with the neuropathological finding of widespread amyloid deposition.^{95,96}

The possible relationship between PD phenotypes—specifically, tremor-dominant versus non-tremor-dominant PD—and classical CSF biomarkers of PD has also been assessed. Recently, reduced levels of A β ₄₂ or an increased t-tau:A β ₄₂ ratio was reported in non-tremor-dominant PD compared with tremor-dominant PD,^{83,97,98} suggesting a more-severe neurodegenerative process in the non-tremor-dominant subtype.

Other CSF biomarkers that are possibly related to AD pathology have also been investigated in patients with PD. CSF levels of neprilysin, a membrane-bound presynaptic protein involved in A β clearance, were unaltered in PD and controls, but were substantially decreased in DLB and PDD. CSF neprilysin activity correlated positively with CSF A β ₄₂ in all groups.⁹⁹ Levels of the inflammatory chemokine fractalkine have also been assessed in the CSF of patients with PD. The fractalkine:A β ₄₂ ratio was closely associated with disease severity.⁶³ Similarly, increased CSF levels of complement C3 or factor H, together with decreased CSF levels of A β ₄₂, have been found to correlate with PD motor severity and cognitive impairment, suggesting that both neuroinflammation and altered A β ₄₂ metabolism are involved in PD progression.¹⁰⁰

Lysosomal enzymes

Aggregation of monomeric α -syn into fibrillar or soluble oligomeric forms, and the formation of Lewy bodies in the PD brain are indicative of protein degradation

impairment. Macroautophagy is the most important pathway for degradation of long-lived proteins such as α -syn (Figure 1), so this pathway could have a central role in PD pathogenesis.^{101–103} During macroautophagy, double-membrane organelles, called autophagosomes, capture a large part of the cytosol and fuse to form a lysosome. An increased number of autophagosomes are observed in postmortem brains of PD patients compared with controls¹⁰⁴ and in cultured cells treated with neurotoxins that specifically target dopaminergic neurons.^{105–107}

Accumulation of autophagosomes is probably not due to induction of autophagy in pathological conditions, but rather to a PD-specific impairment of lysosomal activity.¹⁷ In agreement with this hypothesis, no significant reduction in lysosomal activity has been observed in AD—a disease in which the involvement of cellular stress is well established.^{108,109} A direct relationship between PD pathology and lysosomal dysfunction suggests the potential utility of measures of lysosomal activity as a biomarker for PD.

The role of lysosomal dysfunction in PD pathogenesis is also suggested by the increased risk of parkinsonism in patients with Gaucher disease^{110–112}—a rare lysosomal storage disorder caused by genetic deficiency of the lysosomal enzyme β -glucocerebrosidase (GCase).¹¹³ Several studies have shown a higher prevalence of mutations in the *GBA* gene, which encodes GCase, in patients with PD (2.3–9.4%) compared with the general population (approximately 1%).^{15,114–118} Generalized dystonia associated with akinetic–rigid Parkinsonism has been observed in patients with GM1 gangliosidosis, a lysosomal storage disorder caused by a β -galactosidase deficiency.¹¹⁹ Moreover, lysosomal α -D-galactosidase activity¹²⁰ and gene expression¹²¹ are decreased in leukocytes in patients with sporadic PD compared with controls, and GCase activity in the substantia nigra is similarly reduced in PD,¹⁸ further confirming lysosomal involvement in PD pathogenesis.

Interactions between α -syn and GCase trigger aggregation and neurodegeneration.¹²² Functional loss of GCase induces accumulation of α -syn, thereby provoking aggregation-dependent pathways that lead to neurotoxicity. Moreover, glucosylceramide, the substrate of GCase, facilitates α -syn aggregation and o- α -syn stabilization. Conversely, α -syn aggregates inhibit GCase activity in normal neurons.¹²² These results reinforce a bidirectional pathogenic interaction between GCase and α -syn in synucleinopathies.

According to the above observations, lysosomal system dysfunction represents a target for discovery of potential PD biomarkers. The activity of GCase and other lysosomal hydrolases was measured in the CSF of PD patients¹²³ and other neurodegenerative conditions including DLB.¹²⁴ GCase activity was significantly reduced in PD and DLB patients compared with neurological controls. The potential utility of CSF measures of lysosomal enzyme activity as a biomarker for PD is currently under investigation in two cohorts of patients with PD.

Box 1 | Current issues for CSF biomarkers in Parkinson disease

Criteria for an ideal diagnostic marker and applicability to PD

Detects a fundamental molecular feature of pathogenesis and/or neuropathology

- α -Syn is the major component of Lewy bodies²⁴
- α -Syn lysosomal degradation pathway is altered in PD, with subsequent accumulation, hyperaggregation and formation of neurotoxic oligomers¹²³
- α -Syn interacts with tau and A β , inducing reciprocal hyperaggregation⁹
- DJ-1 is involved in protection against oxidative stress during neurodegeneration⁶²

Validated in neuropathologically confirmed cases

- Total and oligomeric α -syn in the PD brain (but not in CSF)
- Total α -syn measured in CSF of neuropathologically confirmed PD cases⁴⁰

Sensitivity and specificity >80%: unmet need

Detects the disease in the prodromal phase: unmet need

Reliable, non-invasive, simple to perform and inexpensive: unmet need

Status of CSF biomarkers in PD

- Total α -syn and α -syn oligomers versus healthy controls and other neurodegenerative disorders
- α -syn:tau ratio versus other neurodegenerative disorders
- A β_{42} correlates with cognitive impairment
- β -Glucocerebrosidase activity versus healthy controls and other neurodegenerative disorders
- DJ-1: conflicting results

Technical issues

Need to check for blood contamination in the CSF by erythrocyte counting (low cut-off <100–500 cells/ml) or methods to detect haemoglobin products

Specific assay for different oligomeric species: unmet need

Reproducibility between different laboratories: no studies so far, unmet need

Abbreviations: A β , amyloid- β ; CSF, cerebrospinal fluid; PD, Parkinson disease; α -syn, α -synuclein.

Conclusions and perspectives

A modern and pragmatic approach to early diagnosis of neurodegenerative diseases includes imaging and biochemical biomarkers for supporting clinical criteria.² Currently, PD diagnosis relies solely on clinical criteria and is made, therefore, when most dopaminergic neurons have undergone degeneration. This late motor phase is preceded by a prodromal, nonmotor phase, for which a reliable diagnostic marker is currently lacking (Box 1).¹²⁵ Recent data strongly suggest that a combination of CSF biomarkers reflecting converging pathogenic mechanisms (Figure 1) might enable earlier and more accurate diagnosis.

CSF α -syn species—including t- α -syn, o- α -syn, p129- α -syn and/or truncated forms—in combination with classical AD CSF biomarkers, such as A β_{42} , t-tau and p-tau, are promising PD biomarkers because these species have various, interacting roles in pathogenesis. Such an interplay might explain why these peptides, when measured separately, show low specificity as PD biomarkers. DJ-1 represents another potential CSF biomarker in PD, but more studies are needed.

Lysosomal dysfunction is an early event in PD pathogenesis. Accordingly, CSF biomarkers reflecting such dysregulation could be useful for PD diagnosis. In this context, measurement of CSF lysosomal enzymes activity—if confirmed in larger studies—could be a promising approach.

Besides early diagnosis, prediction of cognitive decline in PD represents an unmet clinical need. Several studies have suggested a role for AD pathology in the pathogenesis of cognitive impairment in PD, and baseline

CSF levels of A β ₄₂ might represent a predictive marker for development of cognitive impairment in PD.⁸⁴ Recently, combination of CSF biomarkers linked to neuroinflammation and neurodegeneration has been shown to correlate with disease severity, yielding a first promising result.⁶³

As disease-modifying therapies are likely to be available in future, biomarkers to facilitate early diagnosis and timely initiation of pharmacological treatment, as well as assessment of drug effectiveness, need to be developed. In addition, biomarkers could be used to predict the risk that a given patient will develop adverse drug effects, such as levodopa-induced dyskinesia, psychosis or impulse control disorder. Validation of biochemical assays will be a crucial step in translating CSF biomarkers for PD diagnosis to clinical practice. To achieve this goal, neuropathological validation of biomarkers, which is missing in most PD biomarker studies performed so far, is urgently needed.

At present, a major limitation for the discovery of PD biomarkers is the scarcity of longitudinal studies. Hopefully, the results of the Parkinson Progression Marker Initiative—the largest multicentre longitudinal study to investigate disease biomarkers in early-diagnosed patients with PD and in healthy controls—will lead to a breakthrough in this regard.¹²⁶

In this Review, we considered some CSF candidate biomarkers that have already shown promising results for PD diagnosis and prognosis. Molecules reflecting other important pathogenic mechanisms—such as neuroinflammation, oxidative stress, apoptosis and mitochondrial homeostasis—require further studies to

clarify their potential role as PD biomarkers. Unbiased approaches such as ‘omics’ technologies will be a valuable tool for discovery of new candidates, and recent advances in multiplexed targeted proteomics could represent a breakthrough for validation of candidate biomarkers.^{125,127,128}

Future studies on CSF biomarkers of PD might also generate data regarding the pathophysiological mechanisms of the disease. Indeed, the observation that α -syn induces fibrillization of tau, and that co-incubation of tau and α -syn synergistically promotes fibrillization of both proteins,¹⁰ is of particular relevance for studies investigating these two molecules as putative CSF biomarkers of PD. Similarly, evidence that clearance of oligomeric α -syn intermediates is mediated via the lysosomal degradation pathway¹²⁹ provides a strong rationale for future studies to analyse lysosomal enzymatic activity in the CSF, and the link between these enzymes and α -syn-related molecules in early-stage PD and during disease progression. The complex and dynamic interaction of multiple pathogenic mechanisms in PD necessitates further assessment of combined measurement of several CSF biomarkers in large cohorts, for both early diagnosis and prediction of disease progression.

Review criteria

The PubMed database was searched for papers using the terms “CSF biomarkers”, “alpha-synuclein”, “lysosomal enzymes”, “glucocerebrosidase”, “tau”, “beta-amyloid”, “DJ-1”, and “Parkinson”. The reference lists of identified papers were used as an additional source. Papers published until October 2012 were considered.

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Author contributions

L. Parnetti and A. Castrioto researched data for and wrote the article. L. Parnetti, A. Castrioto, D. Chiasserini, E. Persichetti, O. El-Agnaf and P. Calabresi made substantial contributions to discussion of the article content. L. Parnetti, D. Chiasserini, E. Persichetti, N. Tambasco, O. El-Agnaf and P. Calabresi contributed to review and/or editing of the manuscript before submission.

Supplementary information is linked to the online version of the paper at www.nature.com/nrneuro.